# QUANTUM TRAFFIC GRIDLOCK

Quantum Optimization for Coordinated Urban Traffic Signals | IBM Qiskit Fall Fest 2026 | QC@IITI · IIT Indore

## 1. BACKGROUND

Traffic intersections influence one another: allowing vehicles through one junction can increase queues downstream. Develop a quantum optimization workflow that coordinates traffic-light phases across a simplified network. The emphasis is on mathematical modelling, QUBO construction, QAOA implementation, and honest experimental validation—not building a realistic traffic simulator.

## 2. CHALLENGE OBJECTIVE

Build a scenario-independent program that accepts a traffic scenario, constructs a justified QUBO (or Ising Hamiltonian), runs QAOA, and returns a signal schedule and its raw sampled distribution. Hidden scenarios will test generalization. No QUBO coefficients, Hamiltonian, penalty weights, or optimization solution will be supplied.

## 3. MANDATORY TRAFFIC MODEL

Two connected intersections, A and B, over four 30-second slots. Each intersection gives green to exactly one phase per slot: 1 = North-South (NS), 0 = East-West (EW). The eight decision bits are ordered A slots 1-4 followed by B slots 1-4. A quadratic surrogate is allowed; auxiliary variables may increase the QUBO size beyond eight.

For each slot: (1) add external arrivals and transfers from the preceding slot; (2) the green approach serves up to its capacity; (3) record delay and overflow from queues remaining after service; (4) on the directed A-to-B link, transfer floor(fraction × departures + 0.5) vehicles from the linked approach into B at the beginning of the next slot. Transfers after the last slot leave the model. Other departures leave the model. Queues start empty. Overflow is penalized, not forbidden; all 256 schedules are valid.

## 4. PUBLIC REFERENCE SCENARIO R

| PARAMETER | VALUE |
|---|---|
| Link | A → B, EW approach, transfer fraction 0.6 |
| Capacity / storage | 5 vehicles served per green approach per slot; nominal queue storage 7 |
| Weights | Overflow 4; switching 0.5 |
| A arrivals (NS / EW) | [3, 6, 6, 3] / [5, 1, 1, 2] |
| B arrivals (NS / EW) | [3, 5, 4, 3] / [2, 6, 4, 6] |

Verification vectors (A bits | B bits → official cost): 1010 | 1010 → 116.0; 1111 | 1111 → 146.0; 0000 | 0000 → 237.0; 0111 | 1001 → 92.5. These are scorer checks, not recommended schedules.

## 5. OFFICIAL COST & SCORER

Minimize cost = delay + overflow_weight × overflow + switch_weight × switches. Delay sums all remaining queues after service across slots (vehicle-slots); overflow sums excess above nominal storage; switches count changes between adjacent slots at each intersection (no pre-slot-1 switch cost). The released scorer.py is the authoritative evaluator. It takes scenario JSON and an eight-bit schedule, but does not construct QUBO or optimize.

## 6. REQUIRED EXECUTABLE PIPELINE

Your repository must support: python run.py --scenario <scenario.json> --seed <int> --out <dir> without manual changes.

| Output | Required content |
|---|---|
| qubo.json | QUBO matrix and offset (or Ising h, J); variable-to-schedule-bit mapping; identify auxiliary variables |
| circuit.qasm | Final bound-parameter QAOA circuit in OpenQASM 3 |
| counts.json | Exactly 4,096 raw final-circuit shots as bitstring → count |
| schedule.json | Decoded eight-bit schedule(s), explicitly documenting bit order |

Limits: at most 16 qubits; QAOA depth $p \leqslant 4$; at most 300 training objective evaluations; exactly 4,096 final shots; deterministic seed; CPU-only and no network during judging; at most 10 minutes per seed on an organizer-specified 8 GB reference environment. Fewer shots may be used during training. Classical refinement is allowed only as a separately labelled hybrid result, not as raw-QAOA sampling output. Judges may check the QUBO, circuit, and counts for consistency.

## 7. HIDDEN SCENARIOS & TIMELINE

Hidden scenarios retain the same traffic model and eight decisions, varying arrivals (0–8), capacity (4–6), storage (5–8), overflow weight (2–6), switch weight (0.25–1.5), and a single A → B link on NS or EW with fraction 0.4–0.8. Teams must not hard-code R. Hour 0: PS, scenario R and scorer released. Hour 18: shared evaluation scenario H released. Hour 24: tagged repository, report and outputs for R and H submitted. After the event: judges run the same tagged code on unseen scenario S.

## 8. QUANTUM SAMPLING QUALITY

For each scenario, judges enumerate all 256 schedules to compute the best official cost C* and uniform-random mean C_rand. E_Q is the mean official cost of the 4,096 raw QAOA samples. Lift = (C_rand - E_Q)/(C_rand - C*). Organizers exclude hidden scenarios where the denominator is too small.

Elite enrichment: select exactly the 13 lowest-cost schedules, breaking ties by canonical ascending bitstring. Enrichment = (fraction of shots in that elite set)/(13/256); random sampling has expected enrichment 1. Robustness: report lift across at least five fixed seeds; judges use the minimum lift clipped to [0,1], avoiding unstable ratios. Final thresholds and seeds will be announced before the challenge begins. Finding a single optimum among 4,096 shots is not sufficient evidence of QAOA performance.

## 9. REQUIRE BASELINES & DIAGNOSTICS

Compare: (1) best of four alternating schedules; (2) per-slot greedy control serving the larger queue after arrivals (ties to NS); (3) exact enumeration of 256 schedules; (4) uniform-random expected cost; (5) the exact minimizer of your own QUBO evaluated by the official scorer (surrogate ceiling); (6) QAOA raw samples and optionally a separately labelled hybrid method. Report on R and H. Quantum advantage is not required or expected at this scale.

## 10. MATHEMATICAL INTEGRITY & AI USE

AI assistants and general-purpose libraries are permitted; declare assistance and independently verify results. Event-specific formulation and code must be developed during the hackathon. A QUBO produced only by enumerating all schedules, finding the optimum and encoding that answer is not an acceptable traffic-model derivation. Enumeration remains permitted as a benchmark. The submitted QUBO must arise from a documented, scenario-independent modelling rule grounded in traffic behaviour. Hard-coded schedules or scenario-specific lookups are prohibited. Team size: 1–4. There is no oral defence; judging uses only submitted code, outputs and written report.

## 11. OPTIONAL EXTENSIONS

Examples: three-intersection chains, minimum-green rules, time-varying arrivals, decomposition or hybrid methods, and noise analysis. Extensions require a precise mathematical definition and reproducible evidence; scale alone earns no credit.

## 12. DELIVERABLES

Submit a public GitHub repository containing the runnable pipeline, instructions, and outputs for R and H; a technical report of at most three pages explaining modelling, QUBO derivation, QAOA design, benchmarks and limitations; and a one-page AI-use disclosure outside the report limit. An optional video (up to three minutes) may supplement the submission but is not scored. No interviews or oral presentations.

## 13. EVALUATION RUBRIC (100 marks)

| Criterion | Marks | Assessment |
|---|---|---|
| Traffic formulation and QUBO | 25 | Documented model/derivation (15); surrogate regret relative to true optimum (10), full at $\leqslant 5\%$, zero at $\geqslant 25\%$ |
| QAOA implementation and verification | 20 | Reproducible execution (8); circuit/QUBO/count consistency (6); depth, optimizer and resource choices (6) |
| Hidden-scenario sampling quality | 25 | Lift (12), elite enrichment (8), clipped worst-seed robustness (5); published calibrated thresholds |
| Generalization and robustness | 15 | Unmodified execution on H and S (9); ablations over depth, penalties and initialization (6) |
| Innovation and extensions | 10 | Relevance, mathematical justification, implementation and experimental evidence |
| Report and reproducibility | 5 | Clear derivation, setup, experiments, limitations and AI disclosure |

Judging is entirely submission-based. Organizers will inspect the tagged code, rerun the pipeline, compare the official scores, and evaluate the written report. Incorrect or non-reproducible quantum results cannot earn full quantum-performance credit. Performance thresholds and judging hardware must be frozen and published before the event.